

Airbnb City Analysis Dashboard Using Power BI

Submitted By

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Project Title

Airbnb City Analysis Dashboard Using Power BI

Dataset Name

Airbnb Listings Dataset

Dataset Description

The Airbnb Listings Dataset contains detailed information about Airbnb properties from Chicago and New Orleans. The dataset includes information related to room types, availability, neighborhoods, bathrooms, bedrooms, and listing details.

The dataset was previously analyzed during Exploratory Data Analysis (EDA) and was later used for dashboard creation and visualization using Microsoft Power BI.

Dashboard Objective

The primary objective of this dashboard is to compare Airbnb listings across Chicago and New Orleans and identify trends related to listing distribution, room types, availability, property characteristics, and neighborhood popularity.

The dashboard aims to support data-driven decision-making through visual analytics and interactive filtering.

Tools Used

- Microsoft Power BI Desktop
 - Power Query Editor
 - Airbnb Listings Dataset
 - GitHub
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Data Cleaning and Transformation

The following preprocessing and transformation steps were performed before visualization:

- Imported Airbnb datasets into Power BI.
- Added city labels for Chicago and New Orleans datasets.
- Combined datasets using Append Queries.
- Removed inconsistencies and verified data quality.
- Organized and prepared fields for dashboard creation.
- Created a unified dataset for comparative analysis.

These transformation steps improved data quality and enabled meaningful visual analysis.

Dashboard Visualizations

Visual 1: City-wise Listings Count

Description

This visualization compares the number of Airbnb listings available in Chicago and New Orleans.

Insight

Chicago contains a larger number of Airbnb listings than New Orleans, indicating a stronger Airbnb market presence.

Insert Screenshot Here

Visual 2: Listings Distribution by City (Pie Chart)

Description

This pie chart shows the percentage distribution of Airbnb listings between both cities.

Insight

Chicago contributes a larger share of total listings, while New Orleans contributes a smaller portion.

Insert Screenshot Here

Visual 3: Room Type Distribution

Description

This visualization shows the distribution of Airbnb room types.

Insight

Entire Home/Apartment is the most common room type, followed by Private Rooms.

Insert Screenshot Here

Visual 4: Average Availability by City

Description

This chart compares average availability across cities.

Insight

Availability patterns vary slightly between Chicago and New Orleans, providing insights into booking behavior.

Insert Screenshot Here

Visual 5: Price Distribution by Room Type

Description

This chart shows listing distribution across room types and pricing-related characteristics.

Insight

Different room types contribute differently to the Airbnb marketplace.

Insert Screenshot Here

Visual 6: Bedrooms Distribution

Description

This chart displays the distribution of listings based on bedroom count.

Insight

Most Airbnb listings contain fewer bedrooms, indicating a larger number of compact accommodations.

Insert Screenshot Here

Visual 7: Bathrooms Distribution

Description

This chart displays the distribution of listings according to bathroom count.

Insight

One-bathroom properties dominate the dataset, while larger properties occur less frequently.

Insert Screenshot Here

Visual 8: Neighborhood Analysis

Description

This visualization highlights listing distribution across neighborhoods.

Insight

Certain neighborhoods have significantly higher Airbnb activity compared to others.

Insert Screenshot Here

Visual 9: Interactive City Filter (Slicer)

Description

An interactive filter allows users to switch between Chicago and New Orleans.

Insight

Users can perform city-specific analysis and compare trends dynamically.

Insert Screenshot Here

Key Findings

1. Chicago has a higher number of Airbnb listings than New Orleans.
 2. Entire Home/Apartment is the most preferred room type.
 3. Smaller properties dominate the Airbnb marketplace.
 4. Availability patterns differ across cities.
 5. Certain neighborhoods attract higher Airbnb activity.
 6. Interactive filtering improves user experience and dashboard usability.
 7. Property characteristics such as bedrooms and bathrooms influence listing distribution.
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Business Impact Summary

The dashboard provides useful insights for Airbnb hosts, analysts, and stakeholders.

The analysis helps identify:

- Popular property types.
- High-demand neighborhoods.

- Listing availability patterns.
- City-wise Airbnb market performance.
- Property distribution trends.

These insights can support pricing strategies, investment planning, and operational decision-making.

Dashboard Screenshots

Insert all project screenshots in the following order:

1. Dataset Import
2. Chicago Dataset Transformation
3. New Orleans Dataset Transformation
4. Append Queries Process
5. Combined Dataset with City Column
6. City-wise Listings Count Chart
7. Listings Distribution by City (Pie Chart)
8. Room Type Distribution
9. Average Availability by City
10. Bedrooms Distribution
11. Bathrooms Distribution
12. Neighborhood Analysis
13. City Filter (Slicer)

Conclusion

The Airbnb City Analysis Dashboard successfully transformed raw Airbnb data into meaningful business insights using Microsoft Power BI. Through data cleaning, transformation, interactive filtering, and visual storytelling, the dashboard provides a comprehensive comparison of Airbnb performance across Chicago and New Orleans.

The project demonstrates practical knowledge of Power BI, dashboard design principles, data visualization techniques, and business intelligence concepts while supporting data-driven decision-making.